



Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

College of Computer Studies

ITEC 204 – Data Structures and Algorithms

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- I. **Objective:** Write a short explanation of what your program is designed to do. (Minimum of 3 sentences)

Answer: The program is designed as an IT Automation Incident Ticket Manager that helps manage and organize incident tickets generated by automation bots, where each ticket contains important information such as the incident ID, the bot involved, and a short description of the problem. This system aims to make the management of incident tickets easier by allowing the user to add, display, search, remove, and count the records stored in the list, so the user can easily keep track of the active incidents. It also aims to apply the concept of a linear data structure in a simple IT-related situation by storing the incident tickets in a list and managing each record one by one through the different operations provided by the system.

- II. **Program Operations:** Briefly explain how your program performs each operation. (Insert Specific Code Line and Your Explanation)

Add Incident:

Specific Code:

```
incident_id = input("Enter Incident ID: ")
bot = input("Enter Bot: ")
description = input("Enter Short Description: ")

ticket = IncidentTicket(incident_id, bot, description)
self.tickets.append(ticket)
```

Explanation: These lines are used to get the Incident ID, Bot, and Short Description from the user. The information is then used to create a new incident ticket, and self.tickets.append(ticket) adds the new ticket to the list of active tickets.



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Display Incidents:

Specific Code:

```
if len(self.tickets) == 0:  
    print("No active incident tickets.")  
else:  
    for ticket in self.tickets:
```

Explanation: These lines are used to check if there are any tickets in the list. If the list is empty, the program shows that there are no active tickets. If there are tickets, the for loop goes through each ticket so its information can be displayed.

Search Incident:

Specific Code:

```
search_id = input("Enter Incident ID to search: ")  
  
for ticket in self.tickets:  
    if ticket.incident_id == search_id:
```

Explanation: These lines are used to get the Incident ID that the user wants to search for. The for loop checks each ticket in the list, while the if statement compares the ticket's Incident ID with the ID entered by the user to find a matching ticket.

Remove Resolved Incident:

Specific Code:

```
remove_id = input("Enter Incident ID to remove: ")  
  
for ticket in self.tickets:  
    if ticket.incident_id == remove_id:  
        self.tickets.remove(ticket)
```

Explanation: These lines are used to get the Incident ID of the ticket that the user wants to remove. The program checks the tickets one by one, and if the ID matches, self.tickets.remove(ticket) removes that ticket from the active ticket list.



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Count Active Incidents:

Specific Code:

```
def count_tickets(self):  
    print("Total number of active tickets:", len(self.tickets))
```

Explanation: These lines are used to count the active incident tickets in the list. The `len(self.tickets)` gets the total number of tickets, and the print statement displays the result to the user.

- III. **Flowchart:** Create a flowchart showing the overall process and logic of your program.
- IV. **Conclusion:** Briefly explain what you learned from the activity and how the linear data structure was applied.

In this activity, I learned how to use a linear data structure to store and manage incident tickets in a program. I learned how to add, display, search, remove, and count tickets using a list. The `self.tickets` list was used to keep the incident tickets in order, and I also learned how the program can access each ticket one by one when searching or displaying them. This activity helped me understand how a linear data structure can be used in a simple IT support system.

